NGUYEN DUY TRANG

Open to Opportunities

Mobile Engineer | Flutter | Native iOS & Android

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PROFESSIONAL SUMMARY

OVERVIEW

Mobile Engineer with 5 years of production experience across Flutter, iOS, and Android, directly contributing to 1M+ download social network ecosystems, multi-vertical POS/Kiosk hardware systems (F&B, Retail, Spa Services), and cutting-edge Vision AI products. Strong command of Feature-First Clean Architecture, offline-first architectures, granular Role-Based Access Control (RBAC), 60 FPS rendering optimization with zero memory leaks, and deep peripheral hardware integration (ESC/POS thermal printers, barcode scanners). Pioneered Google Gemini Multimodal Vision AI and Flutter Generative UI (A2UI Protocol).

VERIFIED PRODUCTION HIGHLIGHTS

METRICS

- **1M+ Downloads:** Operating large-scale social travel network & real-time OTT on iOS & Android.
- **-30% Tx Time:** Optimized ordering, billing & checkout on multi-vertical POS & self-order Kiosks.
- **99.5%+ Crash-Free:** Production reliability, strict RAM lifecycle management, and locked 60 FPS.
- **Offline-First & RBAC:** SQLite order sync; permission boundaries for Cashier, Warehouse, Manager.
- **Realtime & Hardware:** Low-latency WebSocket chat/calls, ESC/POS thermal printers & barcode scanners.
- **198 Tests 100%:** Comprehensive automated test suite, Fastlane & GitHub Actions CI/CD pipelines.

TECHNICAL SKILLS MATRIX

COMPETENCIES

Languages & Core:	Flutter (Dart), Swift, SwiftUI, UIKit, Kotlin, TypeScript.
Architecture & State:	Clean Architecture, Riverpod 2.x, BLoC, Provider, MVVM, SOLID, Modularization, Offline-First, RBAC.
Real-Time, Media & Data:	WebSocket, Socket.IO, RESTful APIs, SQLite, Local Caching & Sync, iOS CallKit.
Hardware & POS Modules:	Platform Channels, Thermal POS (ESC/POS), Barcode Scanner, Warehouse Management, Multi-tier Pricing.
AI Tools, CI/CD & QA:	Cursor IDE, Claude Code, Fastlane, GitHub Actions, Bitrise, Git Worktrees, Scrum/Agile.

FEATURED FLAGSHIP PROJECT

PERSONAL ARCHITECTURE

ASTROBITE — AI Food Scanner & GenUI Calorie/Macro Tracker / AI Coach

2026 - Present

Mobile Engineer & System Architect — Intelligent Nutrition & Dietary Vision Assistant

Tech Stack: Flutter 3.x, Dart 3.x, Riverpod 2.6 (Code-Gen, AsyncNotifier), AutoRoute 9.x, Google Gemini 2.0/3.8 Flash (firebase_ai), iOS WidgetKit & Android AppWidget (home_widget), Apple Health & Health Connect, Fastlane, GitHub Actions CI/CD (198 Unit/Widget Tests).

- Architected AstroBite (v2.0.0) based on **Feature-First Clean Architecture** with Riverpod 2.x code-gen across 10 decoupled business modules.
- Engineered camera scanner utilizing **Gemini Multimodal Vision AI (firebase_ai)** recognizing complex meals in **2–4 seconds**, estimating calories and macros (Carbs, Fat, Protein) with **0 memory leaks**.
- Pioneered **Flutter Generative UI (GenUI) Chat Cockpit** via A2UI protocol and Gemini 3.8 Flash, dynamically rendering interactive widgets (MealQuickLogCard, MacroBudgetGauge) with 1-tap Firestore logging.
- Developed **Cross-Platform OS Home Widgets** (iOS WidgetKit & Android AppWidget via home_widget) supporting instant camera scanner deep-link **< 300ms** and reactive data sync with zero background battery drain.
- Established a suite of **198 automated unit & widget tests (100% passing)** with zero flaky tests; automated multi-platform release builds and deployment using Fastlane & GitHub Actions.

NGUYEN DUY TRANG — COMMERCIAL WORK EXPERIENCE & CREDENTIALS

5+ YEARS HANDS-ON EXPERIENCE

COMMERCIAL WORK EXPERIENCE

TRACK RECORD

FINVIET TECHNOLOGY

06/2025 – Present

Mobile Developer — Multi-Industry POS Kiosk Ecosystem (F&B, Retail & Spa Services)

Tech Stack: Flutter, Android Kotlin, BLoC, SQLite, Platform Channels, ESC/POS Thermal Printer, Barcode Scanner, RBAC, Fastlane.

- Architected **POS & self-order Kiosk** checkouts tailored for 3 business verticals: **F&B** (table layout, kitchen ticket printing), **Retail** (barcode scanning, rapid cart), and **Spa Services** (technician booking, treatment packages), cutting **30%** transaction time.
- Designed granular **Role-Based Access Control (RBAC)** across distinct operational roles (Cashier, Warehouse Keeper, Spa Technician, Store Manager, Admin).
- Developed **Goods & Multi-tier Pricing** module: custom pricing per branch, spa treatment packages, F&B food combos, and automatic promotional rules.
- Engineered **Warehouse Management** subsystem: stock-in/stock-out receipts, inter-branch transfers, and real-time inventory tracking.
- Built comprehensive **Reporting & Analytics**: shift revenue breakdown, multi-model sales statistics (F&B, Retail, Spa), and stock movement reconciliation.
- Architected **Offline-first** order sync (BLoC, SQLite) and developed custom **Platform Channels (Kotlin)** interfacing with ESC/POS thermal receipt printers and hardware barcode scanners.
- Configured automated build and release pipelines using **Fastlane & GitHub Actions**.

HAHALOLO SOCIAL TRAVEL NETWORK

05/2020 – 06/2025 (5 years)

Mobile Developer — Hahalolo Social Travel Network & Halome OTT App

Tech Stack: iOS Swift, SwiftUI, UIKit, Flutter Add-to-App, WebSocket, Socket.IO, iOS CallKit, Firebase Crashlytics, Bitrise, Fastlane.

- Scalably operated and maintained flagship social mobile application surpassing **1,000,000+ production downloads (1M+)** across iOS and Android.
- Engineered **Halome** real-time messaging subsystem from scratch, supporting chat, media sharing, and low-latency audio/video calls via WebSocket & Socket.IO.
- Developed native iOS modules with **Swift, SwiftUI, UIKit** and seamlessly integrated them into Flutter via **Flutter Add-to-App**.
- Integrated Firebase FCM with **iOS CallKit** for lock-screen incoming calls; sustained a **99.5%+ crash-free session rate** via Crashlytics.
- Optimized memory lifecycles, widget rebuilds, and render cascades, locking smooth **60 FPS** performance even on low-end hardware.
- Automated end-to-end TestFlight testing and App Store/Google Play distribution pipelines using Bitrise and Fastlane.

EDUCATION & CREDENTIALS

ACADEMIC

FPT POLYTECHNIC COLLEGE
Associate Degree in Information Technology (Software Development Major)

English: Professional Working • **Vietnamese:** Native

ENGINEERING DISCIPLINE & METHODOLOGY

PRACTICES

Architecture & Quality: Feature-First Clean Architecture, Strict SOLID principles, Ponytail Mindset (Zero Bloat, shortest working diff).

Testing & Delivery: 100% Deterministic Testing (Domain, Data, Widget), Automated CI/CD (Fastlane, GitHub Actions, Bitrise).